

# Neural Networks Essentials

## Self-Supervised Learning (SimCLR)

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# Outline

Introduction to Self-Supervised Learning

Contrastive Learning

SimCLR Framework

Experimental Evaluation

Discussion and Conclusion

# Self-Supervised Learning Overview

- ▶ Learning representations from unlabeled data
- ▶ Uses intrinsic structure of data to create pseudo-labels
- ▶ Avoids costly manual annotation
- ▶ Two main paradigms: **generative** (modeling pixel distribution) and **discriminative** (pretext tasks)

# Motivation for Self-Supervision

- ▶ Scarcity of labeled data in many real-world applications
- ▶ Ability to pretrain large models on massive unlabeled datasets
- ▶ Improved downstream performance on classification, detection, and segmentation tasks

# Contrastive Learning Principles

- ▶ Learn representations by contrasting **positive** and **negative** pairs
- ▶ **Positive pair**: two different views (augmentations) of the same image
- ▶ **Negative pairs**: different images or non-corresponding augmentations
- ▶ **Objective**: maximize agreement between positive pairs while separating negatives

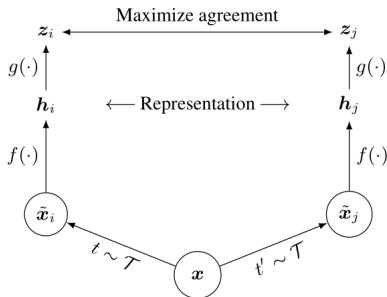
# NT-Xent Loss

- ▶ NT-Xent (Normalized Temperature-scaled Cross Entropy) loss is used
- ▶ Cosine similarity is computed between projection vectors
- ▶ Temperature parameter  $\tau$  scales the similarities

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}$$

# Overview of SimCLR

- ▶ **Data Augmentation:** Critical for creating effective positive pairs
- ▶ **Encoder:** Standard architectures (e.g., ResNet) to extract features
- ▶ **Projection Head:** A small MLP that maps representations to a latent space where contrastive loss is applied
- ▶ **Contrastive Loss:** NT-Xent loss is applied on the projected features



# Data Augmentation in SimCLR

- ▶ **Random Cropping and Resizing:** Ensures different parts of the same image are compared
- ▶ **Color Distortion:** Prevents reliance on low-level color cues
- ▶ **Gaussian Blur:** Further diversifies image views

**Key Insight:** The composition of these augmentations defines a challenging pretext task that forces the model to learn robust features



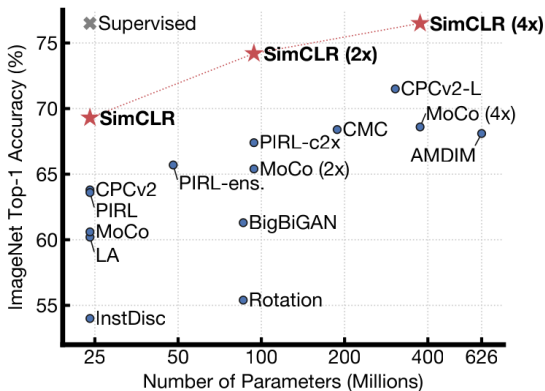
# Encoder and Projection Head

- ▶ **Encoder:** Typically a ResNet (e.g., ResNet-50) without modifications
- ▶ **Projection Head:** A nonlinear MLP (often with one hidden layer and ReLU) improves the quality of the representation
- ▶ **Observation:** The representation from the encoder (before projection) is more effective for downstream tasks

# Training Considerations

- ▶ **Large Batch Sizes:** More negative samples per batch improve performance
- ▶ **Longer Training:** Benefits contrastive learning more than supervised learning
- ▶ **Normalization:**  $L_2$  normalization is key to effective contrastive learning

# Performance on ImageNet



# Performance on ImageNet

- ▶ **Linear evaluation protocol:** Train a linear classifier on frozen features
- ▶ SimCLR achieves competitive accuracy compared to supervised baselines
- ▶ Benefits observed in semi-supervised settings (with limited labeled data)

**Result Example:** A ResNet-50 pretrained with SimCLR achieved up to 76.5% top-1 accuracy

# Comparison with State-of-the-Art

- ▶ SimCLR outperforms previous self-supervised and semi-supervised methods
- ▶ Maintains comparable or superior performance to supervised learning with sufficient pretraining
- ▶ Versatile transfer learning performance on multiple downstream tasks

# Key Takeaways

- ▶ The right data augmentations are crucial in self-supervised contrastive learning
- ▶ A simple framework (without memory banks or specialized architectures) can achieve state-of-the-art results
- ▶ Nonlinear projection heads help to decouple the contrastive loss from the final representation used in downstream tasks
- ▶ Larger batch sizes and longer training times significantly benefit the contrastive approach

# Conclusion

- ▶ SimCLR represents a powerful, yet conceptually simple, framework for self-supervised learning
- ▶ It emphasizes the importance of strong data augmentations and thoughtful architectural choices
- ▶ The framework bridges the gap between self-supervised and supervised methods, enabling efficient learning from unlabeled data